Seongkwang Kim

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Professional Experience

- Assistant Professor, School of Cybersecurity, Korea University, Seoul, Korea — Mar. 2026 - Present
- Senior Engineer in Samsung SDS, Seoul, Korea — Mar. 2022 - Feb. 2026
  - Research on MPCitH/VOLEitH-based signature
    - Main contributor in AIMer team (NIST R1 candidate, KpqC selected)
    - Vector semi-commitment applied to MPCitH/VOLEitH-based signature schemes
  - Research on authenticated encryption with associated data (AEAD)
    - Design and security proof of efficient beyond-birthday-bound AEAD
    - Design and security proof of nonce-misuse resistant AEAD
  - Research on privacy-preserving protocols
    - Privacy-preserving record linkage using circuit-based private set intersection
    - Oblivious pseudorandom function based on oblivious key-value store

Education

- Ph.D. in Information Security — Mar. 2018 - Feb. 2022
  - Where: KAIST, Daejeon, Korea
  - Advisor: Jooyoung Lee
  - Research Area: HE-friendly ciphers, transciphering framework, provable security
- M.Sc. in Mathematical Science — Mar. 2016 - Feb. 2018
  - Where: KAIST, Daejeon, Korea
  - Advisor: Sanggeun Han
  - Research Area: cryptanalysis of LWE
- B.Sc. in Mathematics — Mar. 2012 - Feb. 2016
  - Where: POSTECH, Pohang, Korea

Publications

Authors are listed in alphabetical order by last name, unless an asterisk(*) is indicated. Daggers (†) indicate co-first authors.

Academic Papers

1. W. Chung, S. Hwang, **S. Kim**, B. Lee, and J. Lee. "Extending Accordion Security for Tighter Reduction to AE Security". The 32nd Annual International Conference on the Theory and Application of Cryptology and Information Security (ASIACRYPT 2026), to appear.
2. **S. Kim**, B. Lee, and M. Son. "Shorter VOLE-in-the-Head-based Signatures from Vector Semi-Commitment". The 33rd ACM Conference on Computer and Communications Security (CCS 2026), to appear.
3. W. Chung, S. Hwang, **S. Kim**, B. Lee, and J. Lee. "Security of NIST SP800-90A Hash-based DRBGs in the Seedless Model". The 11th IEEE European Symposium on Security and Privacy (EuroS&P 2026).
4. W. Chung, S. Hwang, **S. Kim**, B. Lee, and J. Lee. "Making GCM Great Again: Toward Full Security and Longer Nonces". The 44th Annual International Conference on the Theory and Applications of Cryptographic Techniques (EUROCRYPT 2025).
5. **S. Kim**, B. Lee, and M. Son. "Relaxed Vector Commitment for Shorter Signatures". The 44th Annual International Conference on the Theory and Applications of Cryptographic Techniques (EUROCRYPT 2025).
6. K. Han, **S. Kim**, and Y. Son. "Private Computation on Common Fuzzy Records". Proceedings on Privacy Enhancing Technologies Symposium (PoPETs 2025).
7. K. Han, **S. Kim**, B. Lee, and Y. Son. "Revisiting OKVS-based OPRF and PSI: Cryptanalysis and Better Construction". The 30th Annual International Conference on the Theory and Application of Cryptology and Information Security (ASIACRYPT 2024).
8. *__S. Kim__[†], J. Ha[†], M. Son, B. Lee, D. Moon, J. Lee, S. Lee, J. Kwon, J. Cho, H. Yoon, and J. Lee. "AIM: Symmetric Primitive for Shorter Signatures with Stronger Security". The 30th ACM Conference on Computer and Communications Security (CCS 2023).
9. J. Ha, **S. Kim**, B. Lee, J. Lee, and M. Son. "Rubato: Noisy Ciphers for Approximate Homomorphic Encryption". The 41st Annual International Conference on the Theory and Applications of Cryptographic Techniques (EUROCRYPT 2022).
10. J. Cho, J. Ha, **S. Kim**, B. Lee, J. Lee, J. Lee, D. Moon, and H. Yoon. "Transciphering Framework for Approximate Homomorphic Encryption". The 27th Annual International Conference on the Theory and Application of Cryptology and Information Security (ASIACRYPT 2021).
11. *J. Ha, **S. Kim**, W. Choi, J. Lee, D. Moon, H. Yoon, and J. Cho. "Masta: An HE-friendly Cipher Using Modular Arithmetic". IEEE Access 10.1109/ACCESS.2020.3033564, 2020.
12. **S. Kim**, B. Lee, and J. Lee. "Tight Security Bounds for Double-Block Hash-then-Sum MACs". The 39th Annual International Conference on the Theory and Applications of Cryptographic Techniques (EUROCRYPT 2020).

Preprints

- S. Kim, B. Lee, and M. Son. "Shorter VOLE-in-the-Head-based Signatures from Vector Semi-Commitment". Cryptology ePrint Archive. Report 2025/1077. 2025. https://eprint.iacr.org/2025/1077.

Technical Report

- *J. Lee, J. Cho, J. Ha, **S. Kim**, J. Kwon, B. Lee, J. Lee, S. Lee, D. Moon, M. Son, and H. Yoon. “The AIMer Signature Scheme (Ver. 2.1)”. 2024. https://aimer-signature.org.
- *J. Lee, J. Cho, J. Ha, **S. Kim**, J. Kwon, B. Lee, J. Lee, S. Lee, D. Moon, M. Son, and H. Yoon. “The AIMer Signature Scheme (Ver. 2.0)”. Submission to Korean Post-Quantum Cryptography (KpqC) Competition 2nd Round. 2024. https://aimer-signature.org.
- ***S. Kim**, J. Ha, M. Son, and B. Lee. “Efficacy and Mitigation of the Cryptanalysis on AIM”. The 5th NIST PQC Standardization Conference. 2024. https://eprint.iacr.org/2023/1474.
- ***S. Kim**, J. Cho, M. Cho, J. Ha, J. Kwon, B. Lee, J. Lee, J. Lee, S. Lee, D. Moon, M. Son, and H. Yoon. “The AIMer Signature Scheme (Ver. 1.0)”. Submission to NIST Call for Additional Signature Schemes. 2023. https://aimer-signature.org.
- ***S. Kim**[†], J. Ha[†], M. Son, B. Lee, D. Moon, J. Lee, S. Lee, J. Kwon, J. Cho, H. Yoon, and J. Lee. “The AIMer Signature Scheme (Ver. 0.9)”. Submission to Korean Post-Quantum Cryptography (KpqC) Competition. 2022. https://aimer-signature.org.

Ph.D. Dissertation

- **S. Kim**. "On Homomorphic Encryption, Transciphering Frameworks, and HE-friendly Ciphers". 2022. KAIST.

Repositories

- https://github.com/KAIST-CryptLab/RtF-Transciphering RtF framework with HERA, Rubato
- https://github.com/samsungsds-opensource/AIMer AIMer

Talks and Posters

- “Hash-based Signature”. DGIST AI & Crypto Summer Camp 2026. Jul. 2026. Daegu, Korea.
- “The KpqC Competition”. Invited Talk, GlobalPlatform Security Task Force Meeting. Apr. 2026. Bangkok, Thailand.
- “The AIMer Signature Scheme”. KpqC Winter Camp 2026. Feb. 2026. Seoul, Korea.
- “Relaxed Vector Commitment for Shorter Signatures”. Invited Talk, ICISC 2025. Nov. 2025. Seoul, Korea.
- “AIMer Standard Format”. 2025 KpqC Conference. Nov. 2025. Seoul, Korea.
- “Hash-based Signatures”. KIISC Cryptography Education Program. Nov. 2025. Online.
- “AIMer Standardization Document Writing Plan”. 2025 KpqC Workshop. Jul. 2025. Taean, Korea.
- “Secure Multiparty Computation, MPC-in-the-Head, and AIMer”. KIISC Cryptography Education Program. Jul. 2025. Online.
- “AIMer v2.1 and Beyond”. Cryptography Research Society (in KIISC) 2025 Mid-term Workshop. Jun. 2025. Busan, Korea.

- “AIMer v2.1 and Beyond”. Invited Talk, KMS Spring Meeting 2025. Apr. 2025. Daejeon, Korea.
- “Revisiting OKVS-based OPRF and PSI: Cryptanalysis and Better Construction”. Asiacrypt 2024. Dec. 2024. Kolkata, India.
- “Updates on AIMer”. KpqC 9th Workshop. Oct. 2024. Seoul, Korea.
- “Circuit-PSI and Applications”. NIST Workshop on Privacy Enhancing Cryptography. Sep. 2024. Online.
- “AIMer”. KpqC Contest Colloquium. Aug. 2024. Seoul, Korea.
- (Poster) “The AIMer Signature Scheme”. NIST 5th PQC Standardization Conference. Apr. 2024. Rockville, MD, US. https://csrc.nist.gov/Events/2024/fifth-pqc-standardization-conference.
- “Efficacy and Mitigation of the Cryptanalysis on AIM”. NIST 5th PQC Standardization Conference. Apr. 2024. Rockville, MD, US.
- “AIM: Symmetric Primitive for Shorter Signatures and Stronger Security”. ACM CCS 2023. Nov. 2023. Copenhagen, Denmark.
- “The AIMer Signature Scheme”. Invited Talk, 2nd Oxford PQC Summit. Sep. 2023. Oxford, United Kingdom. https://www.maths.ox.ac.uk/events/conferences/past-events/oxford-post-quantum-cryptography-workshop-2023.
- “Signature Schemes based on the MPC-in-the-Head Paradigm”. Invited Talk, Ewha-KMS International Cryptography Workshop 2023. Jul. 2023. Seoul, Korea.
- “Reducing the Overhead of Approximate Homomorphic Encryption”. Invited Talk, KMS Fall Meeting 2022. Oct. 2022. Seoul, Korea.
- “Rubato: Noisy Ciphers for Approximate Homomorphic Encryption”. Eurocrypt 2022. Jun. 2022. Trondheim, Norway. https://youtu.be/TE_sYzJtZQc (in English).
- “Transciphering Framework for Approximate Homomorphic Encryption”. Invited Talk, CryptoLab. Dec. 2021. Seoul, Korea.
- “Transciphering Framework for Approximate Homomorphic Encryption”. Asiacrypt 2021. Dec. 2021. Online. https://youtu.be/r3_07fWqOas (in English).
- “Transciphering Framework for Approximate Homomorphic Encryption”. Security @ KAIST. Nov. 2021. Online. https://youtu.be/xKEgtZeMTaw?t=6434 (in Korean).
- “Hybrid Framework for Approximate Computation over Encrypted Data”. KMS Spring Meeting 2021. Apr. 2021. Online.

Patents

- K. Han, **S. Kim**, B. Lee, Y. Song, J. Seo, and I. Hwang. “Method, apparatus and computer program for bypassing verification in homomorphic encryption”. KR1020250193693. Dec. 2025.
- **S. Kim**, K. Han, and B. Lee. “Method for Multiple Applications of an Oblivious Pseudo-Random Function Protocol between a Receiving Device and a Sending Device based on an Oblivious Key-Value Store Algorithm, and Device using the Same”. KR1020250091679. Jul. 2025.

- B. Lee and **S. Kim**. “Method, Apparatus, System and Computer Program for Generating Variable-Output-Length Pseudo-Random Function based on Block Cipher”. KR1020250059321. May. 2025.
- **S. Kim**, B. Lee, M. Son. “Method, Apparatus, System and Computer Program for Zero-Knowledge Proof based on Binary Tree”. (KR1020250156598, US19189453, EP25172274). Feb. 2025.
- Y. Son, **S. Kim**, B. Lee, and K. Han. “Method for implementing a private set intersection protocol using an oblivious pseudorandom function based on minicrypt, and a terminal device using the same”. (KR1020250109577, US19005316). Jan. 2024.
- Y. Son, K. Han, **S. Kim**. “Method for Generating Common Identifier and Apparatus Therefor”. (KR1020240153894, US18396442, EP232189712). Sep. 2023.
- **S. Kim**, D. Moon, J. Kwon, S. Lee, J. Lee, M. Son, B. Lee, and J. Ha. “Method for Calculating using an One-Way Function Efficient in a Zero Knowledge Proof, and Apparatus Implementing the Same Method". (KR1020240073510, US18387520, EP232031583). Nov. 2022.
- Y. Son, K. Han, **S. Kim**. “Method for Protecting Data Based on Private Set Union Protocol, and Apparatus Implementing the Same Method”. (KR1020240029490, US18216223, EP231846056). Oct. 2022.
- D. Moon, J. Lee, J. Lee, Y. Son, **S. Kim**, J. Ha, M. Son, and B. Lee. “Method for Calculating Using a Zero Knowledge Proof-Friendly One-Way Function, and Apparatus Implementing the Same Method”. (KR1020230161195, US18198667, EP231728221, SG10202301388U). May 2022.
- J. Lee, D. Moon, H. Yoon, J. Cho, **S. Kim**, J. Lee, and J. Ha. “Method and Apparatus for Generating Key Stream”. (KR1020220146115, US17514135). Apr. 2021.
- J. Lee, D. Moon, H. Yoon, J. Cho, E. Kim, **S. Kim**, J. Lee, J. Ha, W. Choi. “Apparatus and Method for Encryption, Apparatus and Method for Converting Ciphertext”. (KR20210129573, US17081862). Oct. 2020.
- D. Moon, H. Yoon, and J. Cho, **S. Kim**, J. Lee, J. Ha, and W. Choi. “Apparatus and Method for Symmetric-key Cryptography”. KR1020220022826. Oct. 2020.

Skills

- I speak Korean as a native and English fluently as a second language.
- Familiar with C/C++ (with x86 intrinsics), Python, Mathematica and LaTeX
- Proficient to use PET libraries (e.g. SEAL, HElib, LibOTe)

Teaching Experiences

- Regular lecture (Korea University)
  - SCS503 Modern cryptography: Spring 2026
  - SMRT205 Mathematics for AI security: Spring 2026
- Counseling assistant (KAIST): Sep. 2016 - Feb. 2021
- Teaching assistant (KAIST)
  - IS511 Information security: Spring 2018, Spring 2019
  - CS204 Discrete mathematics: Fall 2019, Spring 2021

Services

Program committee:

- DPM Workshop 2026
- ISC 2025, 2026
- ICISC 2023, 2024, 2025, 2026

External reviewer:

- Eurocrypt 2023, 2026
- Asiacrypt 2019, 2020, 2024
- PKC 2024
- ICISC 2018
- ProvSec 2018

Honors and Awards

- The final selected digital signature algorithm (AIMer) in KpqC Competition (2025)
- The bronze award at Samsung Paper Award in 2023, 2025
- The best dissertation award at Korean Mathematical Society in 2023
- The 3rd award at iDash Competition (Track 4: Secure Record Linkage) in 2022
- The 2nd award at Korea Cryptography Contest (hosted by Korea Cryptography Forum) in 2018, 2024

Other Experiences

- Exchange student at NUS, Singapore in 2015
- Cellist in POSTECH orchestra (Mar. 2012 - Feb. 2016) / KAIST orchestra (Mar. 2016 - Feb. 2020)
- I have traveled to ...
  - Asia: JP, CN, TW, MY, ID, VN, SG, TH, LA, MM, IN, NP, MV, MN, KZ, IL
  - Europe: NO, CZ, PL, DE, NL, CH, IT, FR, ES, UK, DK, AT, VA, RU
  - America & Oceania & Africa: US (MD, NY, DC, GU), NZ, MA

Previous Email Addresses

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